

## Lesson Plan

|                                          |                                                                                                                                                       |                                    |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Name of the Programme:</b>            | <b>B.Tech. (Hons.)</b>                                                                                                                                |                                    |
| <b>Title of the Course:</b>              | <b>Operating System</b>                                                                                                                               |                                    |
| <b>Course code</b>                       | <b>CS1103</b>                                                                                                                                         |                                    |
| <b>Semester:</b>                         | <b>II</b>                                                                                                                                             |                                    |
| <b>Names of the Course Instructor(s)</b> | <b>Prof. Priyanka S, Dr. Debasish Mukherjee, Dr. Ishita Chakraborty, Dr. Srinivas Mishra, Prof. Aisiri K S, Dr. V. Balaji , Dr. Suruchi Sabherwal</b> |                                    |
| <b>Course credit</b>                     | <b>No. of hours per week (2+0+2)</b>                                                                                                                  | <b>Total no. of Teaching hours</b> |
| <b>03</b>                                | <b>04 Hours</b>                                                                                                                                       | <b>60 Hours</b>                    |

| Session | Module No. | Topic                                  | Pedagogy / Activities             | Pre-reading / Reference |
|---------|------------|----------------------------------------|-----------------------------------|-------------------------|
| 1.      | I          | Introduction & Role of OS              | Interactive lecture, discussion   | Silberschatz Ch-1       |
| 2.      |            | OS Types                               | Concept mapping & examples        | Stallings Ch-2          |
| 3.      |            | OS Functions & Services                | Block-diagram activity            | Silberschatz Ch-1       |
| 4.      |            | System Structures & Abstract View      | Chalk-talk, architecture diagrams | Silberschatz Ch-2       |
| 5.      |            | System Calls                           | Demo using Linux syscall examples | TLPI Ch-4               |
| 6.      |            | Virtual Machines & Case Studies        | Case-based learning               | Silberschatz Ch-2       |
| 7.      |            | Linux Commands & Shell Programs        | Hands-on terminal practice        | Petersen Linux          |
| 8.      |            | Linux Commands & Shell Programs        | Hands-on terminal practice        | Petersen Linux          |
| 9.      |            | Command Line Arguments & Env Variables | Code-run-verify                   | TLPI Ch-6               |
| 10.     |            | Lab Reinforcement & Output Validation  | Peer demonstration, reflection    | Lab Manual              |

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|-----|----|------------------------------------------|---------------------------|--------------------|
| 11. |    | OS & System Calls – Revision             | Concept recap, Q&A        | Silberschatz Ch-1  |
| 12. |    | Linux & Shell Programming – Practice     | Command repetition        | Petersen Linux     |
| 13. | II | Process Concepts & PCB                   | Diagram activity, Q&A     | Silberschatz Ch-3  |
| 14. |    | Inter-Process Communication              | Whiteboard examples       | Stallings Ch-3     |
| 15. |    | Threads & Multithreading                 | Analogy-based explanation | Silberschatz Ch-4  |
| 16. |    | Scheduling Basics & Metrics              | Worksheet problems        | Silberschatz Ch-5  |
| 17. |    | FCFS, SJF, Priority Scheduling           | Numerical exercises       | Stallings Ch-5     |
| 18. |    | Round Robin & Algorithm Comparison       | Discussion + examples     | Silberschatz Ch-5  |
| 19. |    | fork(), exec(), wait(), Process Creation | Guided coding             | TLPI               |
| 20. |    | Thread Creation using pthreads           | Hands-on implementation   | TLPI – Pthreads    |
| 21. |    | FCFS & SJF Scheduling Simulation         | Test-case analysis        | Lab Manual         |
| 22. |    | RR & Priority Scheduling Simulation      | Output interpretation     | Lab Manual         |
| 23. |    | CPU Scheduling – Numerical Practice      | FCFS, SJF, RR problems    | Silberschatz Ch-5  |
| 24. |    | Scheduling – Lab Reinforcement           | Performance comparison    | Lab Manual         |
| 25. |    | Unit I & II Revision                     | Concept recap             | Textbook summaries |
| 26. |    | Synchronization & Race Conditions        | Real-world analogies      | Silberschatz Ch-6  |

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|-----|-----|----------------------------------------|---------------------------|---------------------|
| 27. | III | Semaphores                             | Guided problem solving    | Stallings Ch-6      |
| 28. |     | Monitors                               | Pseudocode walkthrough    | Silberschatz Ch-6   |
| 29. |     | Hardware Synchronization               | Mini-lecture & examples   | Stallings Ch-6      |
| 30. |     | Deadlock Concepts & Conditions         | Scenario-based discussion | Silberschatz Ch-7   |
| 31. |     | Deadlock Handling & Avoidance          | Banker's concept examples | Stallings Ch-7      |
| 32. |     | IPC using pipe()                       | Run-observe-explain       | TLPI – IPC          |
| 33. |     | Producer-Consumer with Semaphores      | Coding practice           | Silberschatz Ch-6   |
| 34. |     | Banker's Algorithm Program             | Execution & validation    | Lab Manual          |
| 35. |     | Deadlock – Numerical Practice          | Banker's safety analysis  | Silberschatz Ch-7   |
| 36. |     | Synchronization – Lab Reinforcement    | Semaphore validation      | Lab Manual          |
| 37. |     | Unit III Revision                      | Logic & diagram revision  | Silberschatz Ch-6,7 |
| 38. | IV  | Memory Management Strategies           | Concept walkthrough       | Silberschatz Ch-8   |
| 39. |     | Contiguous & Non-Contiguous Allocation | Diagram explanation       | Stallings Ch-8      |
| 40. |     | Paging & Address Translation           | Worked examples           | Silberschatz Ch-8   |
| 41. |     | Virtual Memory & Demand Paging         | Q&A + case discussion     | Silberschatz Ch-9   |
| 42. |     | File System Concepts                   | Concept mapping           | Stallings – FS      |
| 43. |     | File System Design & Implementation    | Case illustrations        | Silberschatz Ch-10  |

|     |   |                                          |                          |                     |
|-----|---|------------------------------------------|--------------------------|---------------------|
| 44. |   | FIFO Page Replacement Simulation         | Trace-table evaluation   | Silberschatz Ch-9   |
| 45. |   | LRU Page Replacement Simulation          | Output comparison        | Lab Manual          |
| 46. |   | Memory & Paging Practice Lab             | Guided revision          | Previous outputs    |
| 47. |   | Memory Management – Numerical Practice   | Paging & VM problems     | Silberschatz Ch-8,9 |
| 48. |   | Page Replacement – Lab Reinforcement     | FIFO vs LRU comparison   | Lab Manual          |
| 49. | V | Disk Structure & Management              | Diagram-based lecture    | Silberschatz Ch-10  |
| 50. |   | FCFS Disk Scheduling                     | Numerical activity       | Stallings Ch-10     |
| 51. |   | SSTF Scheduling                          | Problem solving          | Silberschatz Ch-10  |
| 52. |   | SCAN & CSCAN Scheduling                  | Worked examples          | Stallings Ch-10     |
| 53. |   | I/O Systems Overview                     | Guided discussion        | Stallings – I/O     |
| 54. |   | System Protection & Security Basics      | Case-study discussion    | Silberschatz Ch-15  |
| 55. |   | FCFS & SSTF Disk Scheduling Programs     | Coding & interpretation  | Lab Manual          |
| 56. |   | SCAN Scheduling & Performance Comparison | Observation & reflection | Lab Manual          |
| 57. |   | Disk Scheduling – Numerical Practice     | Worked examples          | Silberschatz Ch-10  |
| 58. |   | Disk Scheduling – Lab Reinforcement      | Re-execution & analysis  | Lab Manual          |

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|-----|--|----------------------------|------------------------------|-------------------------|
| 59. |  | Unit IV & V<br>Revision    | Numerical &<br>concept recap | Silberschatz<br>Ch-8–10 |
| 60. |  | Overall Course<br>Revision | discussion                   | Notes                   |

### Assessment and Evaluation

| <b>Continuous Internal Evaluation (CIE) :70%</b> |                                                                   |
|--------------------------------------------------|-------------------------------------------------------------------|
| <b>Components of CIE</b>                         | <b>Weightage of each component</b>                                |
| CIE-1<br>- Quiz<br>- Lab Activity                | <b>20% (20 marks)</b><br>10 marks out of 70<br>10 marks out of 70 |
| CIE-2<br>- Theory                                | <b>25% (25 marks)</b><br>25 marks out of 70                       |
| CIE-3<br>- Lab Activity<br>- MOOC Assignment     | <b>25% (25 marks)</b><br>10 marks out of 70<br>15 marks out of 70 |

| <b>Semester End Exams (SEE): 30%</b> |                 |
|--------------------------------------|-----------------|
| Mode of Exam:                        | Theory          |
| Weightage of exam                    | 30%<br>30 MARKS |

### Course Outcomes- Programme Outcomes Matrix

| <b>Course Outcomes</b> |                                                                                                                      | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------------------|----------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| 1.                     | Understand the structural components and core functionalities of an operating system.                                | 3   | 2   | 2   |     | 2   |     |     |     |     |      |      | 2    |
| 2.                     | Apply process management techniques and scheduling algorithms to ensure efficient execution and resource allocation. | 3   | 3   | 2   |     | 3   |     |     |     | 2   |      |      | 2    |

|    |                                                                                                                                                                                                          |   |   |   |  |   |  |  |  |   |   |  |   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|--|---|--|--|--|---|---|--|---|
| 3. | Apply synchronization techniques and design solutions to prevent or resolve deadlocks in concurrent systems.                                                                                             | 3 | 3 | 2 |  | 2 |  |  |  | 2 |   |  | 2 |
| 4. | Illustrate the memory management strategies, principles of file system design, and security mechanisms for efficient performance, resource utilization, and reliable data handling in operating systems. | 3 | 3 | 2 |  | 2 |  |  |  | 2 | 2 |  | 2 |

Note: High: 3, Medium: 2, Low: 1

**Table-1**

| Practical Execution rubrics (Max: 6 marks) |                                                         |                     |                                                                                                    |                                                                                                 |                                                                                                      |
|--------------------------------------------|---------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Sl. No                                     | Criteria                                                | Measuring Methods   | Excellent                                                                                          | Good                                                                                            | Poor                                                                                                 |
| 1.                                         | <b>Understanding of problem statement<br/>(2 Marks)</b> | <b>Observations</b> | The student exhibits a thorough understanding of the requirements for the problem.<br><b>(2 M)</b> | The student has a sufficient understanding of the requirements for the problem.<br><b>(1 M)</b> | The student does not have a clear understanding of the requirements for the problem.<br><b>(0 M)</b> |

|                                    |                                                                   |                          |                                                                                                                                                                     |                                                                                                                              |                                                                                              |
|------------------------------------|-------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 2.                                 | <b>Execution<br/>(2 Marks)</b>                                    | <b>Observations</b>      | The student demonstrates the execution of the program with optimized code and shows the output with appropriate validations and test cases handled.<br><b>(2 M)</b> | The student demonstrates the execution of the program without optimization of the code and shows the output.<br><b>(1 M)</b> | The student has not executed the program.<br><b>(0 M)</b>                                    |
| 3.                                 | <b>Results and Documentation<br/>(2Marks)</b>                     | <b>Observations</b>      | Documentation with appropriate comments and output cases are noted in data sheets.<br><b>(2 M)</b>                                                                  | Documentation with few comments and few output cases are noted in data sheets.<br><b>(1 M)</b>                               | Documentation with no comments and no output cases are noted in data sheets.<br><b>(0 M)</b> |
| <b>Quiz Rubrics (Max: 4 marks)</b> |                                                                   |                          |                                                                                                                                                                     |                                                                                                                              |                                                                                              |
| <b>Sl. No</b>                      | <b>Criteria</b>                                                   | <b>Measuring Methods</b> | <b>Excellent<br/>(2 Marks)</b>                                                                                                                                      | <b>Good<br/>(1 Mark)</b>                                                                                                     | <b>Poor<br/>(0.5 Marks)</b>                                                                  |
| 1.                                 | <b>Conceptual Understanding of Operating System<br/>(2 Marks)</b> | <b>Quiz</b>              | Thorough explanation of concepts                                                                                                                                    | Adequate explanation with some gaps                                                                                          | Unable to explain tools/concepts clearly                                                     |

|    |                                                            |             |                                                            |                                                                |                                        |
|----|------------------------------------------------------------|-------------|------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------|
| 2. | <b>Application of Problem-Solving Techniques (2 Marks)</b> | <b>Quiz</b> | Describes and justifies the selection of specific services | Identifies applicable methods with minor gaps in justification | Lacks clarity or incorrect application |
|----|------------------------------------------------------------|-------------|------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------|

**Table-2**

| Sl. No. | Criteria                                            | Measuring Methods | Excellent                                                               | Good                             | Satisfactory                                   | Poor                                  |
|---------|-----------------------------------------------------|-------------------|-------------------------------------------------------------------------|----------------------------------|------------------------------------------------|---------------------------------------|
| 1       | <b>Conceptual Understanding (6 Marks)</b>           | <b>Quiz</b>       | Answers show clear and accurate understanding of all key concepts (6 M) | Minor conceptual errors (5 M)    | Basic understanding with noticeable gaps (3 M) | Poor or incorrect understanding (1 M) |
| 2       | <b>Application of Concepts (4 Marks)</b>            | <b>Quiz</b>       | Correctly applies concepts to problem-based questions (4 M)             | Mostly correct application (3 M) | Limited or partially correct application (2 M) | Unable to apply concepts (1 M)        |
| 3       | <b>Analytical &amp; Logical Reasoning (3 Marks)</b> | <b>Quiz</b>       | Demonstrates strong reasoning and justification (3 M)                   | Reasoning mostly logical (2 M)   | Limited analytical ability (1 M)               | Illogical or random answers (0 M)     |
| 4       | <b>Accuracy &amp; Completeness (2 Marks)</b>        | <b>Quiz</b>       | All answers accurate and complete (2 M)                                 | Few minor mistakes (1.5 M)       | Several incorrect/incomplete answers (1 M)     | Mostly incorrect (0 M)                |

**Signature of Faculty**